Methods 8

John Dong

Apologies for the late submission! You can view my repo's commit history here if that helps to verify that this is my own work done independent from the solutions.

1.

a.

$$\text{Provided that :} \qquad \bar{Y}(A=0,B=0)=1, \qquad \bar{Y}(A=1,B=0)=5,$$

$$\bar{Y}(A=0,B=1)=4, \qquad \bar{Y}(A=1,B=1)=7$$

we can hold the following to be true.

$$\begin{aligned}\bar{Y}(A=0,B=0) &= \mathbb{E}(Y_i \mid A=0,B=0)\\ &= \mathbb{E}(\beta_0+\beta_1\cdot 0+\beta_2\cdot 0+\beta_3\cdot 0+\epsilon_i)\\ &= \beta_0+0+0+0+\mathbb{E}(\epsilon_i)\\ &= \beta_0+0=1\\ &\therefore \beta_0=1\end{aligned}$$

$$\begin{aligned}\bar{Y}(A=1,B=0) &= \mathbb{E}(Y_i \mid A=1,B=0)\\ &= \mathbb{E}(\beta_0+\beta_1\cdot 1+\beta_2\cdot 0+\beta_3\cdot 0+\epsilon_i)\\ &= \beta_0+\beta_1+0+0+\mathbb{E}(\epsilon_i)\\ &= 1+\beta_1+0=5\\ &\therefore \beta_1=4\end{aligned}$$

$$\begin{aligned}
\bar{Y}(A=0,B=1) &= \mathbb{E}(Y_i \mid A=0,B=1) \\
&= \mathbb{E}(\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 1 + \beta_3 \cdot 0 + \epsilon_i) \\
&= \beta_0 + 0 + \beta_2 + 0 + \mathbb{E}(\epsilon_i) \\
&= 1 + \beta_2 + 0 = 4 \\
&\therefore \beta_2 = 3
\end{aligned}$$

$$\begin{aligned}
\bar{Y}(A=1,B=1) &= \mathbb{E}(Y_i \mid A=1,B=1) \\
&= \mathbb{E}(\beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot 1 + \beta_3 \cdot 1 + \epsilon_i) \\
&= \beta_0 + \beta_1 + \beta_2 + \beta_3 + \mathbb{E}(\epsilon_i) \\
&= 1 + 4 + 3 + \beta_3 = 7 \\
&\therefore \beta_3 = -1
\end{aligned}$$

Final Answer: $\boxed{\beta_0 = 1, \beta_1 = 4, \beta_2 = 3, \beta_3 = -1}$

b.

In order to get estimates for the four regression coefficients, we need estimates for the four averages above. We have two variables, A and B, which can take values 0 or 1 depending on whether or not the value observed was at a "high" or "low" level for each. β_0, the intercept or "baseline", necessitates that we have a treatment group where $A = 0, B = 0$. Then, by comparing to our baseline, we can get our treatment effects for A and B by changing the level one at a time: one group is $A = 1, B = 0$, and the other $A = 0, B = 1$. This lets us estimate β_1 and β_2. We also have to estimate the interaction effect β_3, so we need treatment group $A = 1, B = 1$. This constructs our full factorial design, where we have 2^2 treatment groups, one for each combination of treatment levels. Then, our observational units are evenly split among the treatment groups.

In summary:

Group	Treatment	Number of units
Group 1	$A = 0, B = 0$	$n = 50$
Group 2	$A = 1, B = 0$	$n = 50$
Group 3	$A = 0, B = 1$	$n = 50$
Group 4	$A = 1, B = 1$	$n = 50$